

RegulonDB

From *E. coli* K-12 to a multi-genomic platform.

RegulonDB has been the international curated reference for bacterial transcriptional regulation since 1991. This deck walks through the prototype of **RegulonDB MG** — the redesign that extends curation to *Salmonella enterica*, *Bacillus subtilis*, and additional bacterial models, integrating experimental evidence with computational predictions, orthology, metabolic context, and high-throughput datasets.

3

SEED ORGANISMS

4,639

GENES · *E. COLI*

335

TFS · *E. COLI*

214

REGULONS · *E. COLI*

~600

CURATED REGULONS CROSS-ORGANISM

MULTI-
GENOMIC
PROTOTYPE

Stakeholder

review

v0.9

prototype

2026 · CCG-

UNAM

WHAT YOU'LL SEE

Home · Gene page (araC) · TF page (LexA) · Regulon (AraC, LexA) · Search · Compare cross-organism · Salmonella
RegulonDB MG · CCG-UNAM ·
2026
· B. subtilis

Transcriptional regulation, organism-aware.

Curated interactions, high-throughput datasets and predictions across bacterial model organisms — with cross-organism context preserved.

🔍 Try a gene (araC), TF (LexA), operon (araBAD), or a pathway...

☐ Search across all organisms

Advanced search

Try:

araC

LexA

AraC regulon

σ factors

LexA across species

EcEscherichia coli

K-12 MG1655 · v12.0 · updated 2026-03-14

4,639

CURATED GENES

335

TRANSCRIPTION FACTORS

2,783

OPERONS

214

REGULONS

NCBI:txid511145

Switch organism →

Browse E. coli

≡

Genes

4,639 entries

Genomic coordinates, products, TUs, context.

📁

Operons

2,783 entries

Co-transcribed gene clusters, TSSs, terminators.

🔗

Regulons

214 entries

Gene sets under a single TF; conditions & sites.

🏠

Transcription factors

335 entries

Families, domains, effectors, target sets.

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Promoters

4,174 entries

σ-factor usage, −10/−35 boxes, TSS evidence.

📄

Datasets HT

47 entries

ChIP-seq · ChIP-exo · RNA-seq · gSELEX · DAP-seq.

Release notes

All releases →

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Multi-organism features

- Context preserved across organism switches when the object exists.
- Ortholog tables on every gene & TF page.
- Side-by-side comparison for TFs and regulons.
- Organism-scoped URLs: /ecoli-k12/gene/araC

For curators

The new curation UI (Citrus) is in beta. Request access →

■ GENE · ESCHERICHIA COLI K-12 MG1655

araC — arabinose operon regulatory protein

Dual regulator of the L-arabinose utilization system; activates the araBAD, araC, araE and araFGH operons in the presence of arabinose and represses araC in its absence.

LOCUS TAG	LENGTH	STRAND	COORDS	OPERON
b0064	879 nt · 292 aa	–	70,387–71,265	araC

Add to set

FASTA

GenBank ↗

Overview

Curation v12 · last updated 2026-03-14

Symbol	araC
Product	AraC · dual transcriptional regulator
Synonyms	b0064
Transcription units	araCp → araC
External IDs	EcoCyc:EG10055 ↗ · UniProt:P0A9E0 ↗ · NCBI Gene:944780 ↗

■ TRANSCRIPTION FACTOR · ESCHERICHIA COLI K-12 MG1655

LexA (gene: *lexA*)

Master repressor of the SOS response to DNA damage. Binds the SOS box of ~40 genes; upon DNA damage it undergoes RecA-stimulated auto-cleavage, derepressing the regulon.

FAMILY	LENGTH	CONSENSUS SITE	TARGETS
LexA repressor, S24 peptidase	202 aa	CTGT-N ₈ -ACAG (SOS box, 20 bp)	14 curated

View regulon →

Compare across organisms

Add TF to set

Export PFM/PWM

Domain architecture

Auto-cleavage architecture typical of LexA / S24 peptidase family



Effectors & signals

What activates / inactivates this TF

EFFECTOR / SIGNAL	MODE	EVIDENCE
RecA* (filament)	inactivating	IDA CURATED
ssDNA-stress signal	indirect	IEP CURATED

Regulated targets

14 interactions · see LexA regulon →

Filter targets...	Only IDA	Only repressed	Add selected to set	Export	CSV	TSV	JSON	GFF3
☐ TARGET	EFFECT	MODE	CONDITION	EVIDENCE				
☐ <i>recA</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>lexA</i>	—	repress	auto	IDA CURATED			binding site ↗	
☐ <i>sulA</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>umuDC</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>uvrA</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>uvrB</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>uvrD</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>ruvAB</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>polB</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>dinB</i>	—	repress	SOS off	IDA CURATED			binding site ↗	
☐ <i>dinI</i>	—	repress	SOS off	IEP CURATED			binding site ↗	
☐ <i>sbmC</i>	—	repress	SOS off	IEP CURATED			binding site ↗	
☐ <i>ftsK</i>	—	repress	SOS off	IC CURATED			binding site ↗	
☐ <i>yebG</i>	—	repress	SOS off	IDA CURATED			binding site ↗	

References

1. Little JW; (1991). Mechanism of specific LexA cleavage. Biochimie 73. PMID:11448770 ↗
2. Fernández de Henestrosa AR et al. (2000). Identification of additional genes belonging to the LexA regulon. Mol Microbiol. PMID:16077032 ↗

■ REGULON

·

ESCHERICHIA COLI K-12 MG1655

AraC regulon

L-arabinose utilization regulon. AraC acts as a dual regulator — repressor without arabinose, activator when bound to arabinose.

REGULATOR	MEMBERS	EFFECT	EVIDENCE
AraC	10 genes	Dual (±)	8 IDA / 2 other

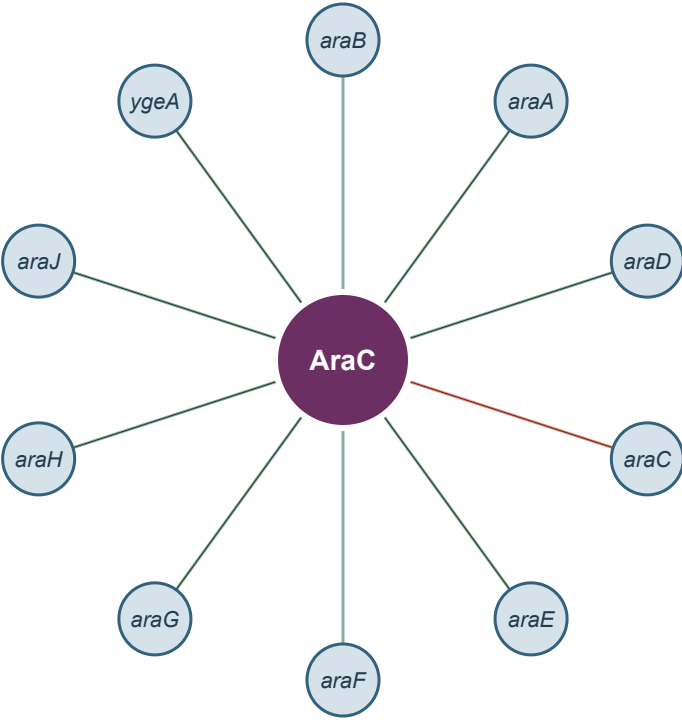
Compare across organisms

Add regulon to set

Download network (SIF)

Network view

Static representation · hover nodes to highlight · open in full visualizer ↗



Legend: ● TF ● target — repress — activate

Members

10 genes

Filter members...						Add selected to set	Export	CSV	TSV	JSON	GFF3
	GENE	OPERON	EFFECT	CONDITION	EVIDENCE						
☐	araB	araBAD	+	+ arabinose	IDA CURATED						
☐	araA	araBAD	+	+ arabinose	IDA CURATED						
☐	araD	araBAD	+	+ arabinose	IDA CURATED						
☐	araC	araC	±	dual (auto)	IDA CURATED						
☐	araE	araE	+	+ arabinose	IDA CURATED						
☐	araF	araFGH	+	+ arabinose	IDA CURATED						
☐	araG	araFGH	+	+ arabinose	IDA CURATED						
☐	araH	araFGH	+	+ arabinose	IDA CURATED						
☐	araJ	araJ	+	+ arabinose	IEP CURATED						
☐	ygeA	ygeA	+	+ arabinose	IEP CURATED						

■ REGULON · *ESCHERICHIA COLI* K-12 MG1655

LexA regulon

SOS response regulon. Repressed by LexA under normal growth; derepressed upon DNA damage via RecA*-stimulated LexA auto-cleavage.

REGULATOR	MEMBERS	EFFECT	EVIDENCE
LexA	14 genes	Repression	11 IDA / 3 other

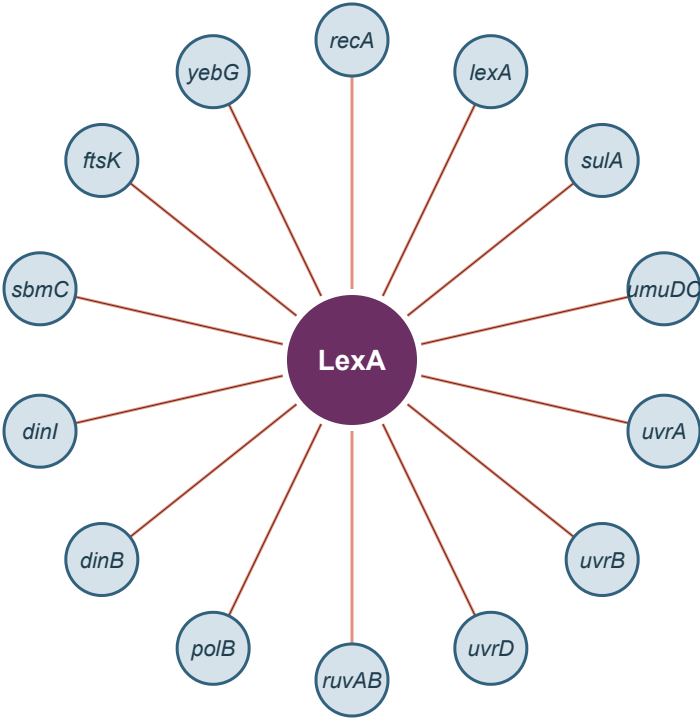
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Members

14 genes

Filter members...					Add selected to set	Export	CSV	TSV	JSON	GFF3
☐	GENE	OPERON	EFFECT	CONDITION	EVIDENCE					
☐	recA	recA	—	SOS off	IDA CURATED					
☐	lexA	lexA	—	auto	IDA CURATED					
☐	sulA	sulA	—	SOS off	IDA CURATED					
☐	umuDC	umuDC	—	SOS off	IDA CURATED					
☐	uvrA	uvrA	—	SOS off	IDA CURATED					
☐	uvrB	uvrB	—	SOS off	IDA CURATED					
☐	uvrD	uvrD	—	SOS off	IDA CURATED					
☐	ruvAB	ruvAB	—	SOS off	IDA CURATED					
☐	polB	polB	—	SOS off	IDA CURATED					
☐	dinB	dinB	—	SOS off	IDA CURATED					
☐	dinI	dinI	—	SOS off	IEP CURATED					
☐	sbmC	sbmC	—	SOS off	IEP CURATED					
☐	ftsK	ftsK	—	SOS off	IC CURATED					
☐	yebG	yebG	—	SOS off	IDA CURATED					

E. coli > Search > “lexA”

Search results

10 matches for “lexA” across 3 organisms

Active filters: q: lexA × ☑️ Cross-organism

OBJECT TYPE	Clear	Sort by	Relevance	Name	Evidence strength	Export	CSV	TSV	JSON	GFF3
☐ gene	4			■ GENE lexA · Escherichia coli K-12 MG1655				CURATED	+ Set	
☐ tf	2				Repressor of the SOS response; auto-cleaves under DNA damage.					
☐ operon	1				📍 NC_000913.3 : 4,254,631 – 4,255,238 (+) RegulonDB: ecoli-k12/gene/lexA					
☐ regulon	1									
☐ promoter	1									
☐ dataset	1									
EVIDENCE	Clear									
☐ curated	8									
☐ predicted	1									
☐ ht	1									
☐ weak	0									
ORGANISM	Clear									
☐ E. coli	7									
☐ S. Typhimurium	1									
☐ B. subtilis	2									
SOURCE	Clear									
☐ RegulonDB curated	9									
☐ High-throughput	2									
☐ Computational	3									
■ TF LexA (DinR) · Bacillus subtilis 168										
					B. subtilis SOS repressor; recognizes Cheo box (CGAAC-N4-GTTCG), distinct from Gram-negative SOS box.					
					RegulonDB: bacillus-subtilis/tf/LexA (DinR)					
■ GENE recA · Escherichia coli K-12 MG1655										
					Recombinase; forms nucleoprotein filaments on ssDNA that stimulate LexA auto-cleavage.					
					RegulonDB: ecoli-k12/gene/recA					
■ PROMOTER lexAp · Escherichia coli K-12 MG1655										
					σ ⁷⁰ promoter upstream of lexA — auto-regulated by LexA itself.					
					RegulonDB: ecoli-k12/promoter/lexAp					
■ OPERON umuDC · Escherichia coli K-12 MG1655										
					SOS-induced operon encoding translesion polymerase V. LexA-repressed.					
					RegulonDB: ecoli-k12/operon/umuDC					
■ GENE lexA · Bacillus subtilis 168										
					Ortholog call inferred from sequence. Functional validation pending.					
					RegulonDB: bacillus-subtilis/gene/lexA					
■ DATASET ChIP-exo LexA (Ishihama 2025) · Escherichia coli K-12 MG1655										
					Genome-wide LexA binding under mitomycin-C stress. 46 peaks; 39 overlap curated targets.					
					RegulonDB: ecoli-k12/dataset/ChIP-exo LexA (Ishihama 2025)					

CROSS-ORGANISM TF

Export comparison (TSV)

Switch TF

LexA across bacterial models

The SOS repressor is structurally conserved from Gram-negatives to Gram-positives, but recognizes **different operator sequences** — classic SOS box in *E. coli* / *Salmonella*, Cheo box in *Bacillus* — and regulates partially overlapping target sets.

Per-organism summary

EcEscherichia coli K-12 MG1655	StSalmonella enterica serovar Typhimurium LT2	BsBacillus subtilis 168
Symbol LexA	Symbol LexA	Symbol LexA (DinR)
Length 202 aa	Length 202 aa	Length 205 aa
Identity to E. coli LexA 100%	Identity to E. coli LexA 91%	Identity to E. coli LexA 37%
Operator CTGT–N₈–ACAG	Operator CTGT–N₈–ACAG	Operator CGAAC–N₄–GTTCG (Cheo box)
Targets (curated) 42	Targets (curated) 38	Targets (curated) 33
Sequence conservation 100%	Sequence conservation 91%	Sequence conservation 37%
Open E. coli page →	Open S. Typhimurium page →	Open B. subtilis page →

Target conservation matrix

Which SOS targets are present in each organism's curated regulon

GENE	E. COLI	S. TYPHIMURIUM	B. SUBTILIS	COVERAGE
STM1254	—	●	—	1/3
dinB	●	●	—	2/3
dinI	●	—	—	1/3
ftsK	●	—	—	1/3
lexA	●	●	●	3/3
polB	●	●	—	2/3
polY1	—	—	●	1/3
polY2	—	—	●	1/3
recA	●	●	●	3/3
ruvAB	●	●	●	3/3
sbmC	●	—	—	1/3
sulA	●	●	—	2/3
umuDC	●	●	—	2/3
uvrA	●	●	—	2/3
uvrB	●	●	—	2/3
uvrBA	—	—	●	1/3
uvrD	●	●	—	2/3
yebG	●	●	—	2/3
yneA	—	—	●	1/3
yneB	—	—	●	1/3
yolD	—	—	●	1/3

Interpretation

Take-aways

- Core SOS response (*recA*, *lexA*, *uvrA/B/D*, *ruvAB*) is conserved across all three.
- Gram-negative–specific targets (*sulA*, *umuDC*) are absent or replaced in *B. subtilis*.
- Operator divergence (SOS box vs Cheo box) reflects *parallel* evolution of the stress response rather than simple drift.

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LexA

AraC regulon

σ factors

LexA across species

StSalmonella enterica

serovar Typhimurium LT2 · v1.4 · updated 2026-02-02

4,556

CURATED GENES

284

TRANSCRIPTION FACTORS

2,491

OPERONS

172

REGULONS

NCBI:txid99287

Switch organism →

Browse S. Typhimurium

Genes
4,556 entries
Genomic coordinates, products, TUs, context.

Operons
2,491 entries
Co-transcribed gene clusters, TSSs, terminators.

Regulons
172 entries
Gene sets under a single TF; conditions & sites.

Transcription factors
284 entries
Families, domains, effectors, target sets.

Promoters
3,120 entries
σ-factor usage, −10/−35 boxes, TSS evidence.

Datasets HT
47 entries
ChIP-seq · ChIP-exo · RNA-seq · gSELEX · DAP-seq.

Release notes

All releases →

v12.0.32026-03-14E. COLI

84 new TF–gene interactions from recent ChIP-exo

Incorporates Ishihama lab 2025 ChIP-exo dataset; 84 new interactions with strong evidence, 19 re-annotated.

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- Organism-scoped URLs: /ecoli-k12/gene/araC

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LexA

AraC regulon

σ factors

LexA across species

BsBacillus subtilis

168 · v1.2 · updated 2026-01-20

4,245

CURATED GENES

258

TRANSCRIPTION FACTORS

2,110

OPERONS

149

REGULONS

NCBI:txid224308

Switch organism →

Browse B. subtilis

≡

Genes

4,245 entries

Genomic coordinates, products, TUs, context.

📁

Operons

2,110 entries

Co-transcribed gene clusters, TSSs, terminators.

🔗

Regulons

149 entries

Gene sets under a single TF; conditions & sites.

🏠

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⇒

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Summary history

[Export data](#)

Evolution of curated transcriptional regulation across releases. Counts reflect cumulative content per organism; cross-organism totals are summed across the most-recent release per organism.

GENES
13,440
unchanged since v12.0.2

TRANSCRIPTION FACTORS
877
unchanged since v12.0.2

REGULONS
535
unchanged since v12.0.2

REG. INTERACTIONS
20,500
+84 since v12.0.2

HT DATASETS
76
+2 since v12.0.2

Major releases

v12.0.3

2026-03-14

84 new TF–gene interactions from recent ChIP-exo

E. COLI

v1.4

2026-02-02

Salmonella Typhimurium LT2 reaches 172 regulons

S. TYPHIMURIUM

v1.2

2026-01-20

B. subtilis sporulation cascade fully curated

B. SUBTILIS

v12.0.2

2025-12-08

GENSOR units for amino-acid biosynthesis

E. COLI

v12.0.1

2025-09-12

gSELEX dataset import (170 conditions)

E. COLI

v1.3

2025-08-19

Two-component systems imported

S. TYPHIMURIUM

v1.1

2025-04-30

SubtiWiki regulator import

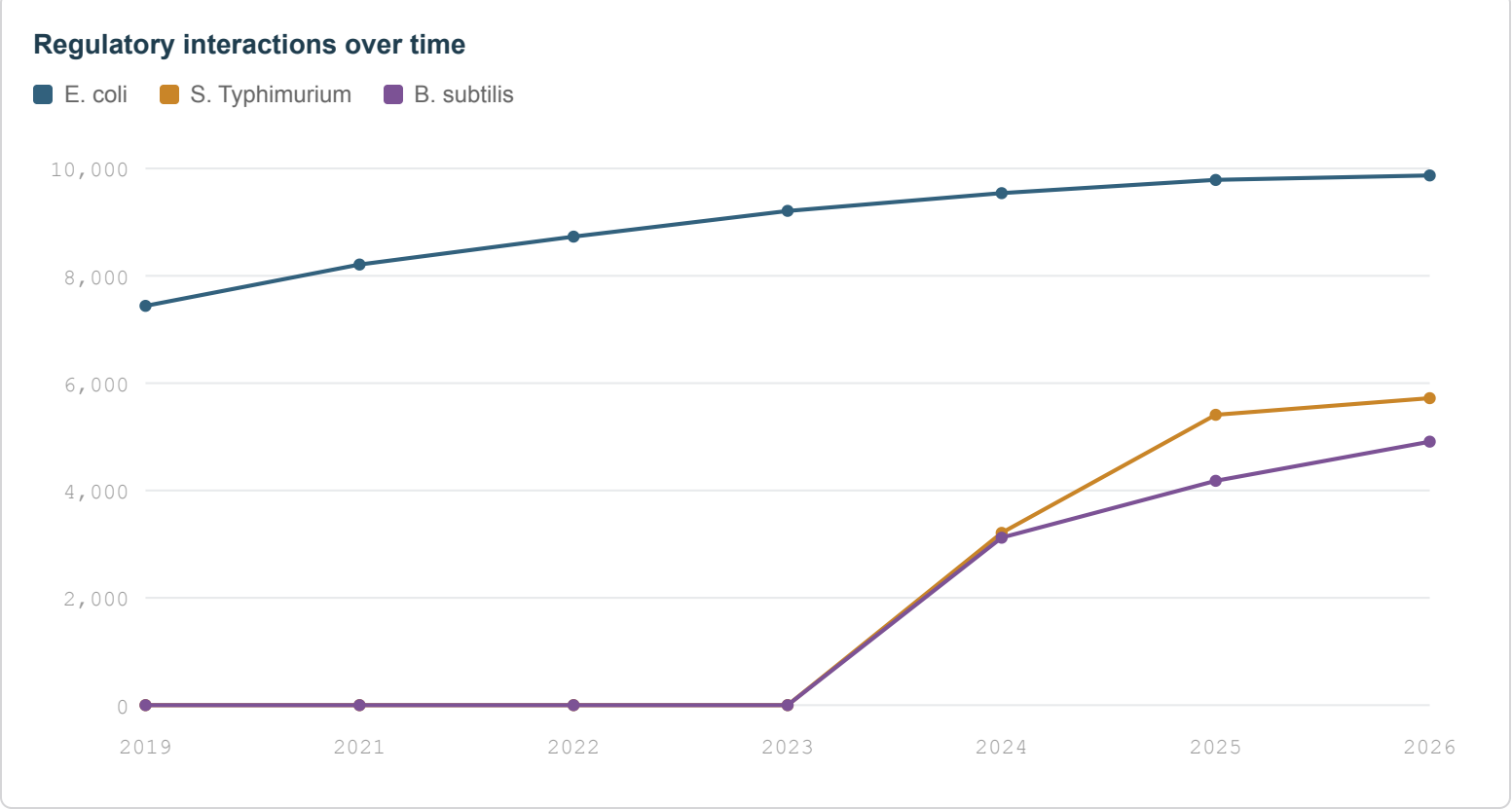
B. SUBTILIS

v1.2

2025-02-10

Initial ChIP-seq import

S. TYPHIMURIUM



About this summary

Cumulative content as of each release across 3 organisms. Counts are computed at curation freeze time; numbers may shift retroactively if past evidence is re-categorized.

Data export

Export CSV

Export TSV

Export JSON

Latest highlight

v12.0.3

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Incorporates Ishihama lab 2025 ChIP-exo dataset; 84 new interactions with strong evidence, 19 re-annotated.

Content changes by release					Export	CSV	TSV	JSON	GFF3
VERSION	RELEASED	ORGANISM	GENES	TFS	REGULONS	INTERACTIONS	TFBSS	HT	
v12.0.3	2026-03-14	<i>E. coli</i>	4,639	335	214	9,870	12,560	47	
v1.4	2026-02-02	<i>S. Typhimurium</i>	4,556	284	172	5,720	6,810	18	
v1.2	2026-01-20	<i>B. subtilis</i>	4,245	258	149	4,910	5,320	11	
v12.0.2	2025-12-08	<i>E. coli</i>	4,639	335	214	9,786	12,498	45	
v12.0.1	2025-09-12	<i>E. coli</i>	4,639	333	213	9,712	12,420	44	
v1.3	2025-08-19	<i>S. Typhimurium</i>	4,548	269	158	5,410	6,320	14	
v1.1	2025-04-30	<i>B. subtilis</i>	4,232	232	124	4,180	4,520	6	
v1.2	2025-02-10	<i>S. Typhimurium</i>	4,532	244	138	4,920	5,640	8	
v12.0	2024-11-04	<i>E. coli</i>	4,636	328	212	9,540	12,090	41	
v1.0	2024-09-15	<i>B. subtilis</i>	4,220	198	80	3,120	3,010	0	
v1.0	2024-06-04	<i>S. Typhimurium</i>	4,520	210	96	3,210	3,480	0	
v11.2	2023-06-21	<i>E. coli</i>	4,632	318	208	9,210	11,680	32	
v11.0	2022-09-08	<i>E. coli</i>	4,621	305	198	8,730	11,140	22	
v10.10	2021-04-15	<i>E. coli</i>	4,598	287	188	8,210	10,620	16	
v10.0	2019-11-22	<i>E. coli</i>	4,540	256	174	7,440	9,810	12	